

Classwork 4

Merit and Excellence problem solving

Grades: [A] [M] [E]

Answers · tutor's working, not an official mark scheme

Show all working. Finish every proof with a sentence.

Part A · Prove, meet, check

A1 Prove that the sum of two consecutive odd numbers is always a multiple of 4. [M]

Two consecutive odd numbers: $2n + 1$ and $2n + 3$

$(2n + 1) + (2n + 3) = 4n + 4 = 4(n + 1)$, which is 4 times a whole number,

so the sum is always a multiple of 4.

A2 Find the points where $y = x^2 - 3x + 1$ meets $y = x - 2$. [A]

$x^2 - 3x + 1 = x - 2$, $x^2 - 4x + 3 = 0$, $(x - 1)(x - 3) = 0$

$x = 1$: $y = -1$; $x = 3$: $y = 1$. Points $(1, -1)$ and $(3, 1)$

A3 Solve $\sqrt{x + 2} = x$. [M]

$x + 2 = x^2$, $x^2 - x - 2 = 0$, $(x - 2)(x + 1) = 0$

Check: $x = 2$ gives $\sqrt{4} = 2$; $x = -1$ gives $\sqrt{1} = 1 \neq -1$, rejected. So $x = 2$.

Part B · A tangent and a calendar

B1 Find k so that the line $y = kx - 1$ is a tangent to $y = x^2 + 3$. [M]

$x^2 + 3 = kx - 1$, $x^2 - kx + 4 = 0$

Tangent: $(-k)^2 - 4(1)(4) = 0$, $k^2 = 16$, $k = 4$ or $k = -4$

B2 On a calendar, a rectangle 2 days wide and 3 weeks tall has top-left number n . Show that the products of its diagonally opposite corners always differ by 14. [M]

Corners: n , $n + 1$ (top right), $n + 14$ (bottom left), $n + 15$ (bottom right)

$(n + 1)(n + 14) - n(n + 15) = n^2 + 15n + 14 - n^2 - 15n = 14$, for every n

Part C · A largest area

C A rectangle has perimeter 24 cm. Show that its area cannot be more than 36 cm².

[E]

Words you may use

width x · quadratic · real solution · discriminant · therefore

Sentence starters (optional): *If the width is x , the length is ... For a real width, ...*

Width x , length $12 - x$, area A : $x(12 - x) = A$, $x^2 - 12x + A = 0$ MP1

A real width needs $b^2 - 4ac \geq 0$: $144 - 4A \geq 0$ MP2

So $A \leq 36$: the area is never more than 36 cm² (the 6 cm square). MP3

MP1 is r; MP2 with the conclusion MP3 is t.

Extension

Where do $x^2 + y^2 = 25$ and $y = x + 1$ meet?

$x^2 + (x + 1)^2 = 25$, $2x^2 + 2x - 24 = 0$, $x^2 + x - 12 = 0$, $x = 3$ or $x = -4$

Points (3, 4) and (-4, -3)